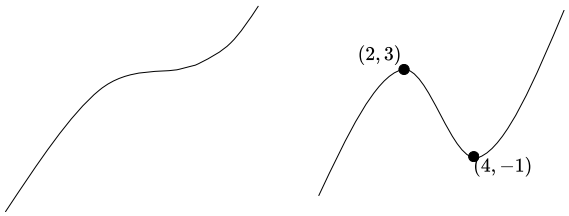
## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 37 · Increasing and Decreasing Functions




A function can also be increasing and decreasing in certain intervals.

What do you think it means for a function to be an ‘increasing function’?

! An increasing function is one whose gradient is always at least 0.

$f'(x) \geq 0$ for all $x$.

We could also write “$f(x)$ is decreasing in the interval $[2, 4]$”

**[a,b]** represents all the real numbers between **a** and **b** inclusive, i.e:

$[a, b] = \{x : a \leq x \leq b\}$

It would be ‘strictly increasing’ if $f(x) > 0$ for all $x$, i.e. is not allowed to go horizontal.

Chapter 12 · Original slide 38 · Examples

Show that the function $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for all real values of $x$.

Tip: To show a quadratic is always positive, complete the square, then indicate the squared term is always at least 0.

Find the interval on which the function $f(x) = x^3 + 3x^2 - 9x$ is decreasing.

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 39 · Test Your Understanding

Show that the function $f(x) = x^3 + 16x - 2$ is increasing for all real values of $x$.

Find the interval on which the function $f(x) = x^3 + 6x^2 - 135x$ is decreasing.

Chapter 12 · Original slide 40 · Exercise 12G – Pearson Pure Y1 page 271

**3** Show that the function $\mathrm{f}(x) = 4 - x(2x^2 + 3)$ is decreasing for all $x \in \mathbb{R}$.

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 41 · Exercise 12G – Pearson Pure Y1 page 271

**4 a** Given that the function $\mathrm{f}(x) = x^2 + px$ is increasing on the interval $[-1, 1]$, find one possible value for $p$.
**b** State with justification whether this is the only possible value for $p$.

Chapter 12 · Original slide 42 · Second Order Derivatives

When you differentiate once, the expression you get is known as the first derivative.

Unsurprisingly, when we differentiate a second time, the resulting expression is known as the second derivative. And so on…

Original Function

First Derivative

Leibniz's

$\frac{d^2y}{dx^2} = 12x^2$

$y\prime = 4x^3$

Newton's

$\ddot{y} = 12x^2$

Lagrange's

$f\prime\prime(x) = 12x^2$

Second Derivative

$\frac{dy}{dx} = 4x^3$

$y = x^4$

$y\prime\prime = 12x^2$

$\dot{y} = 4x^3$

$f(x) = x^4$

$f\prime(x) = 4x^3$

You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon…

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 43 · Just for your interest…

How does the notation $\frac{d^2y}{dx^2}$ work?

Why are the squareds where they are?

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:

$\frac{d}{dx}\left(x^3+1\right) = 3x^2$

Suppose that $y = x^3 + 1$.

Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:

$\frac{d}{dx}\left(x^3+1\right)$

Therefore, if we wanted to differentiate $y$ twice, we'd do:

$\frac{d}{dx}\left(\frac{d}{dx}(y)\right)$
$= \frac{d^2}{dx^2}(y)$
$= \frac{d^2y}{dx^2}$

Chapter 12 · Original slide 44 · Examples

If $y = 3x^5 + \frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 45 · Test Your Understanding

If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

Chapter 12 · Original slide 46 · Exercise 12H – Pearson Pure Y1 page 272

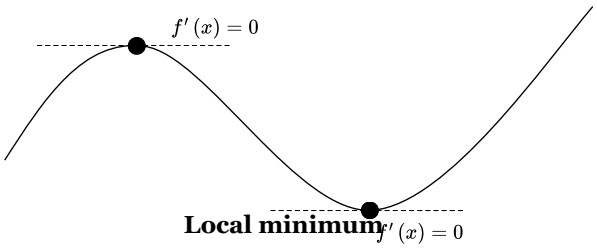
**3** Given that $y = (2x - 3)^3$, find the value of $x$ when $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 47 · Exercise 12H – Pearson Pure Y1 page 272

**4** $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$, $\mathrm{f}''(x) = -1$. Find the value of $p$.

Chapter 12 · Original slide 48 · Stationary/Turning Points




A stationary point is where the gradient is 0, i.e. $f'(x) = 0$.

Local maximum

Note: It's called a 'local' maximum because it's the function's largest output within the vicinity. Functions may also have a 'global' maximum, i.e. the maximum output across the entire function. This particular function doesn't have a global maximum because the output keeps increasing up to infinity. It similarly has no global minimum, as with all cubics.

Find the coordinates of the turning points of $y = x^3 + 6x^2 - 135x$

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 49 · More Examples

Find the least value of

$f(x) = x^2 - 4x + 9$

Method 1: Differentiation

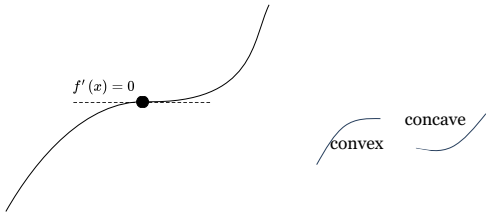
Note: Method 2 is only applicable for quadratic functions. For others, differentiation must be used.

Find the turning point of

$y = \sqrt{x} - x$

Method 2: Completing the square

Chapter 12 · Original slide 50 · Points of Inflection




There's a third type of stationary point (that we've encountered previously):

**Side Note**: Not all points of inflection are stationary points, as can be seen in the example on the right.
A point of inflection which **is** a stationary point is known as a *saddle point*.



A point of inflection is where the curve changes from convex concave (or vice versa).

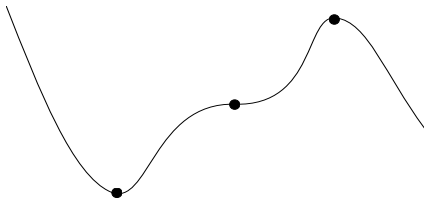
(the same terms used in optics!)

Technically we could label these either way round depending on where we view the curve from. What's important is that the concavity changes.

i.e. the line curves in one direction before the point of inflection, then curves in the other direction after.

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 51 · How do we tell what type of stationary point?



Method 1: Look at gradient just before and just after point.

| Local Maximum | | |
|---|---|---|
| Gradient just before | Gradient at maximum | Gradient just after |
| | | |

| Point of Inflection | | |
|---|---|---|
| Gradient just before | Gradient at p.o.i | Gradient just after |
| | | |

| Local Minimum | | |
|---|---|---|
| Gradient just before | Gradient at minimum | Gradient just after |
| | | |

Chapter 12 · Original slide 52 · How do we tell what type of stationary point?

Method 1: Look at gradient just before and just after point.

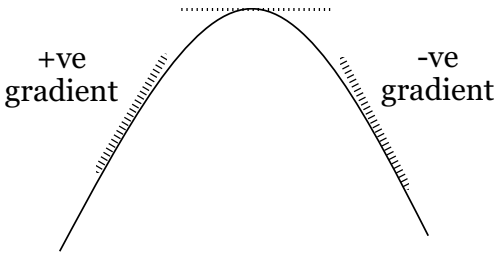
Strategy: Find the gradient for values just before and after $x = 2$. Let's try $x = 1.9$ and $x = 2.1$.

Find the stationary point on the curve with equation $y = x^4 - 32x$, and determine whether it is a local maximum, a local minimum or a point of inflection.

| | **x=1.9** | **x=2** | **x=2.1** |
|---|---|---|---|
| Gradient | | | |
| Shape | | | |

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 53 · Method 2: Using the second derivative




The method of substituting values of $x$ just before and after is a bit cumbersome.

It also has the potential for problems: what if two different types of stationary points are really close together?

Recall the gradient gives a measure of the rate of change of $y$, i.e. how much the $y$ value changes as $x$ changes.

Thus by differentiating the gradient function, the second derivative tells us the rate at what the gradient is changing.

Thus if the second derivative is positive, the gradient is increasing.

If the second derivative is negative, the gradient is decreasing.

At a maximum point, we can see that as $x$ increases, the gradient is decreasing from a positive value to a negative value.

$\therefore \frac{d^2y}{dx^2} < 0$

0

gradient

Chapter 12 · Original slide 54 · Method 2: Using the second derivative

! At a stationary point $x = a$:

• If $f''(a) > 0$ the point is a local minimum.

• If $f''(a) < 0$ the point is a local maximum.

• If $f''(a) = 0$ it could be any type of point, so resort to Method 1.

I will eventually do a 'Just for your Interest...' thingy on why we can't classify the point when $f''(x) = 0$, and how we could use the third derivative!

The stationary point of $y = x^4 - 32x$ is $(2, -48)$. Use the second derivative to classify this stationary point.

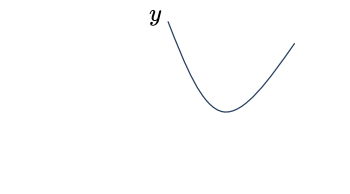
Find $\frac{d^2y}{dx^2}$ for the $x$ at the stationary point.

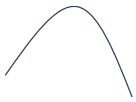
## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 55 · Test Your Understanding

The curve with equation $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point $P$. Use calculus
**(a)** to find the coordinates of $P$, (6)
**(b)** to determine the nature of the stationary point $P$. (3)

Chapter 12 · Original slide 56 · Sketching Graphs



All the way back in Chapter 4, we used features such as intercepts with the axes, and behaviour when $x \to \infty$ and $x \to -\infty$ in order to sketch graphs.

Now we can also find stationary/turning points!

[Textbook] By first finding the stationary points, sketch the graph of $y = \frac{1}{x} + 27x^3$

? As $x \to \infty$, $x \to -\infty$

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 57 · Exercise 12I – Pearson Pure Y1 page 276

**5** By considering the gradient on either side of the stationary point on the curve $y = x^3 - 3x^2 + 3x$, show that this point is a point of inflection. Sketch the curve $y = x^3 - 3x^2 + 3x$.

Chapter 12 · Original slide 58 · Exercise 12I – Pearson Pure Y1 page 276

**6** Find the maximum value and hence the range of values for the function $\mathrm{f}(x) = 27 - 2x^4$.

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 59 · Exercise 12I – Pearson Pure Y1 page 276

**7** $\mathrm{f}(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.
**a** Find the coordinates of the stationary points of $\mathrm{f}(x)$, and determine the nature of each.
**b** Sketch the graph of $y = \mathrm{f}(x)$.

Chapter 12 · Original slide 60 · Extension

**1**
[MAT 2014 1C] The cubic
$y = kx^3 - (k+1)x^2 + (2-k)x - k$
has a turning point, that is a minimum, when $x = 1$ precisely for
A) $k > 0$
B) $0 < k < 1$
C) $k > \frac{1}{2}$
D) $k < 3$
E) all values of $k$

**3**
*[MAT 2001 1E]* The maximum gradient of the curve $y = x^4 - 4x^3 + 4x^2 + 2$ in the range $0 \le x \le 2\frac{1}{5}$ occurs when:
A) $x = 0$
B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$
D) $x = 2\frac{1}{5}$
[STEP I 2007 Q8] A curve is given by:
$y = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$
where $a$ is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) ... when $a = 1$ (iii) when $a = -2$

**4**
Hint: When two curves touch, their $y$ values must match, but what else must also match?

**2**
*[MAT 2004 1B]* The smallest value of the function:
$f(x) = 2x^3 - 9x^2 + 12x + 3$
In the range $0 \le x \le 2$ is what?

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 61 · Solutions to Extension Questions

Chapter 12 · Original slide 62 · Solutions to Extension Questions

## 1.34 Differentiation: Stationary Points

Chapter 12 · Original slide 63 · Solutions to Extension Questions